```
import numpy as np
import pandas as pd
```

```
movies = pd.read_csv('tmdb_5000_movies.csv')
credits = pd.read_csv('tmdb_5000_credits.csv')
```

```
movies.head(1)
```

	budget	genres	homepage	id	keywords	original_language	original_title	overview	popularity	p
0	237000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.avatarmovie.com/	19995	[{"id": 1463, "name": "culture clash"}, {"id":...	en	Avatar	In the 22nd century, a paraplegic Marine is di...	150.437577	

```
credits.head(1)
```

	movie_id	title	cast	crew
0	19995	Avatar	[{"cast_id": 242, "character": "Jake Sully", "...	[{"credit_id": "52fe48009251416c750aca23", "de...

```
movies = movies.merge(credits,on='title')
```

```
movies.head(1)
```

	budget	genres	homepage	id	keywords	original_language	original_title	overview	popularity	p
0	237000000	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	http://www.avatarmovie.com/	19995	[{"id": 1463, "name": "culture clash"}, {"id":...	en	Avatar	In the 22nd century, a paraplegic Marine is di...	150.437577	

1 rows × 23 columns

```
#genres
#id
#keywords
#title
#overview
#cast
#crew
movies = movies[['movie_id','title','overview','genres','keywords','cast','crew']]
```

```
movies.head()
```

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	[{"id": 1463, "name": "culture clash"}, {"id":...	[{"cast_id": 242, "character": "Jake Sully", "...	[{"credit_id": "52fe48009251416c750aca23", "de...
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[{"id": 12, "name": "Adventure"}, {"id": 14, "...	[{"id": 270, "name": "ocean"}, {"id": 726, "na...	[{"cast_id": 4, "character": "Captain Jack Spa...	[{"credit_id": "52fe4232c3a36847f800b579", "de...
2	206647	Spectre	A cryptic message from Bond’s past sends him o...	[{"id": 28, "name": "Action"}, {"id": 12, "nam...	[{"id": 470, "name": "spy"}, {"id": 818, "name...	[{"cast_id": 1, "character": "James Bond", "cr...	[{"credit_id": "54805967c3a36829b5002c41", "de...
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...	[{"id": 28, "name": "Action"}, {"id": 80, "nam...	[{"id": 849, "name": "dc comics"}, {"id": 853,...	[{"cast_id": 2, "character": "Bruce Wayne / Ba...	[{"credit_id": "52fe4781c3a36847f81398c3", "de...

```
movies.isnull().sum()
```

	0
movie_id	0
title	0
overview	3
genres	0
keywords	0
cast	0
crew	0

dtype: int64

```
movies.dropna(inplace=True)
```

```
movies.duplicated().sum()
```

```
np.int64(0)
```

```
movies.iloc[0].genres
```

```
'[{"id": 28, "name": "Action"}, {"id": 12, "name": "Adventure"}, {"id": 14, "name": "Fantasy"}, {"id": 878, "name": "Scienc
```

```
#[{"id": 28, "name": "Action"}, {"id": 12, "name": "Adventure"}, {"id": 14, "name": "Fantasy"}, {"id": 878, "name": "Scienc
```

```
def convert (obj):
  L = []
  for i in ast.literal_eval(obj):
      L.append (i['name'])
  return L
```

```
import ast
ast.literal_eval('[{"id": 28, "name": "Action"}, {"id": 12, "name": "Adventure"}, {"id": 14, "name": "Fantasy"}, {"id": 878
```

```
[{'id': 28, 'name': 'Action'},
 {'id': 12, 'name': 'Adventure'},
 {'id': 14, 'name': 'Fantasy'},
 {'id': 878, 'name': 'Science Fiction'}]
```

```
movies['genres'] = movies['genres'].apply(convert)
```

```
movies.head()
```

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[{"id": 1463, "name": "culture clash"}, {"id":...	[{"cast_id": 242, "character": "Jake Sully", "...	[{"credit_id": "52fe48009251416c750aca23", "de...
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[{"id": 270, "name": "ocean"}, {"id": 726, "na...	[{"cast_id": 4, "character": "Captain Jack Spa...	[{"credit_id": "52fe4232c3a36847f800b579", "de...
2	206647	Spectre	A cryptic message from Bond’s past sends him o...	[Action, Adventure, Crime]	[{"id": 470, "name": "spy"}, {"id": 818, "name...	[{"cast_id": 1, "character": "James Bond", "cr...	[{"credit_id": "54805967c3a36829b5002c41", "de...
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve	[Action, Crime, Drama, Thriller]	[{"id": 849, "name": "dc comics"}, {"id":	[{"cast_id": 2, "character": "Bruce Wayne /	[{"credit_id": "52fe4781c3a36847f81398c3", "de

```
movies['keywords'] = movies['keywords'].apply(convert)
```

```
def convert3 (obj):
  L = []
  counter = 0
  for i in ast.literal_eval(obj):
    if counter != 3:
      L.append (i['name'])
      counter+=1
    else:
      break
  return L
```

```
movies ['cast'] = movies['cast'].apply(convert3)
```

```
movies.head()
```

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[culture clash, future, space war, space colon...	[Sam Worthington, Zoe Saldana, Sigourney Weaver]	[{"credit_id": "52fe48009251416c750aca23", "de...
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drug abuse, exotic island, east india ...	[Johnny Depp, Orlando Bloom, Keira Knightley]	[{"credit_id": "52fe4232c3a36847f800b579", "de...
2	206647	Spectre	A cryptic message from Bond’s past sends him o...	[Action, Adventure, Crime]	[spy, based on novel, secret agent, sequel, mi...	[Daniel Craig, Christoph Waltz, Léa Seydoux]	[{"credit_id": "54805967c3a36829b5002c41", "de...
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve	[Action, Crime, Drama, Thriller]	[dc comics, crime fighter, terrorist, secret	[Christian Bale, Michael Caine, Gary Oldman]	[{"credit_id": "52fe4781c3a36847f81398c3", "de

```
def fetch_director(obj):
  L = []
  for i in ast.literal_eval(obj):
    if i['job'] == 'Director':
      L.append (i['name'])
      break
  return L
```

```
movies['crew'] = movies['crew'].apply(fetch_director)
```

```
movies.head()
```

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[culture clash, future, space war, space colon...	[Sam Worthington, Zoe Saldana, Sigourney Weaver]	[James Cameron]
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drug abuse, exotic island, east india ...	[Johnny Depp, Orlando Bloom, Keira Knightley]	[Gore Verbinski]
2	206647	Spectre	A cryptic message from Bond's past sends him o...	[Action, Adventure, Crime]	[spy, based on novel, secret agent, sequel, mi...	[Daniel Craig, Christoph Waltz, Léa Seydoux]	[Sam Mendes]
3	49026	The Dark Knight Rises	Following the death of District Attorney	[Action, Crime, Drama, Thriller]	[dc comics, crime fighter, terrorist,	[Christian Bale, Michael Caine, Gary Oldman]	[Christopher Nolan]

```
movies['overview'][0] #since this is a string we will convert this into list so that
```

```
'In the 22nd century, a paraplegic Marine is dispatched to the moon Pandora on a unique mission, but becomes torn between f
ollowing orders and protecting an alien civilization.'
```

```
movies['overview'].apply(lambda x:x.split())
```

	overview
0	[In, the, 22nd, century,, a, paraplegic, Marin...
1	[Captain, Barbossa,, long, believed, to, be, d...
2	[A, cryptic, message, from, Bond’s, past, send...
3	[Following, the, death, of, District, Attorney...
4	[John, Carter, is, a, war-weary,, former, mili...
...	...
4804	[El, Mariachi, just, wants, to, play, his, gui...
4805	[A, newlywed, couple's, honeymoon, is, upended...
4806	["Signed,, Sealed,, Delivered", introduces, a,...
4807	[When, ambitious, New, York, attorney, Sam, is...
4808	[Ever, since, the, second, grade, when, he, fi...

4806 rows × 1 columns

dtype: object

```
movies.head()
```

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, Science Fiction]	[culture clash, future, space war, space colon...	[Sam Worthington, Zoe Saldana, Sigourney Weaver]	[James Cameron]
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drug abuse, exotic island, east india ...	[Johnny Depp, Orlando Bloom, Keira Knightley]	[Gore Verbinski]
2	206647	Spectre	A cryptic message from Bond’s past sends him o...	[Action, Adventure, Crime]	[spy, based on novel, secret agent, sequel, mi...	[Daniel Craig, Christoph Waltz, Léa Seydoux]	[Sam Mendes]
3	49026	The Dark Knight Rises	Following the death of District Attorney	[Action, Crime, Drama, Thriller]	[dc comics, crime fighter, terrorist,	[Christian Bale, Michael Caine, Gary Oldman]	[Christopher Nolan]

```
movies['genres'] = movies['genres'].apply(lambda x:[i.replace(" ","") for i in x])
movies['cast'] = movies['cast'].apply(lambda x:[i.replace(" ","") for i in x])
movies['crew'] = movies['crew'].apply(lambda x:[i.replace(" ","") for i in x])
movies['keywords'] = movies['keywords'].apply(lambda x:[i.replace(" ","") for i in x])
```

```
movies.head()
```

	movie_id	title	overview	genres	keywords	cast	crew
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, ScienceFiction]	[cultureclash, future, spacewar, spacecolony, ...	[SamWorthington, ZoeSaldana, SigourneyWeaver]	[JamesCameron]
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drugabuse, exoticisland, eastindiatrad...	[JohnnyDepp, OrlandoBloom, KeiraKnightley]	[GoreVerbinski]
2	206647	Spectre	A cryptic message from Bond’s past sends him o...	[Action, Adventure, Crime]	[spy, basedonnovel, secretagent, sequel, mi6, ...	[DanielCraig, ChristophWaltz, LéaSeydoux]	[SamMendes]
		The Dark Knight	Following the	[Action, Crime	[dccomics,	[ChristianBale,	

```
movies['tags'] = movies['overview'].apply(lambda x:x.split()) + movies['genres'] + movies['keywords'] + movies['cast'] + mo
```

```
movies.head()
```

	movie_id	title	overview	genres	keywords	cast	crew	tags
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...	[Action, Adventure, Fantasy, ScienceFiction]	[cultureclash, future, spacewar, spacecolony, ...	[SamWorthington, ZoeSaldana, SigourneyWeaver]	[JamesCameron]	[In, the, 22nd, century,, a, paraplegic, Marin...
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...	[Adventure, Fantasy, Action]	[ocean, drugabuse, exoticisland, eastindiatrad...	[JohnnyDepp, OrlandoBloom, KeiraKnightley]	[GoreVerbinski]	[Captain, Barbossa,, long, believed, to, be, d...
2	206647	Spectre	A cryptic message from Bond's past sends him o...	[Action, Adventure, Crime]	[spy, basedonnovel, secretagent, sequel, mi6, ...	[DanielCraig, ChristophWaltz, LéaSeydoux]	[SamMendes]	[A, cryptic, message, from, Bond's, past, send...
		The Dark	Following the death of	[Action, Crime	[dccomics, crimefighter	[ChristianBale,		[Following, the, death, of

```
new_df = movies[['movie_id','title','tags']]
```

```
new_df['tags'] = new_df['tags'].apply(lambda x:" ".join(x))
```

```
/tmp/ipython-input-3089450492.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  new_df['tags'] = new_df['tags'].apply(lambda x:" ".join(x))
```

```
new_df.head()
```

	movie_id	title	tags
0	19995	Avatar	In the 22nd century, a paraplegic Marine is di...
1	285	Pirates of the Caribbean: At World's End	Captain Barbossa, long believed to be dead, ha...
2	206647	Spectre	A cryptic message from Bond’s past sends him o...
3	49026	The Dark Knight Rises	Following the death of District Attorney Harve...
4	49529	John Carter	John Carter is a war-weary, former military ca...

```
new_df['tags'][0]
```

```
'In the 22nd century, a paraplegic Marine is dispatched to the moon Pandora on a unique mission, but becomes torn between f
ollowing orders and protecting an alien civilization. Action Adventure Fantasy ScienceFiction cultureclash future spacewar
spacecolony society spacetravel futuristic romance space alien tribe alienplanet cgi marine soldier battle loveaffair antiw
```

```
new_df['tags'] = new_df['tags'].apply(lambda x:x.lower())
```

```
/tmp/ipython-input-3214958533.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  new_df['tags'] = new_df['tags'].apply(lambda x:x.lower())
```

```
import nltk
```

```
from nltk.stem.porter import PorterStemmer
ps = PorterStemmer()
```

```
def stem(text):
  y = []
  for i in text.split():
      y.append(ps.stem(i))
  return " ".join(y)
```

```
new_df['tags'] = new_df['tags'].apply(stem)
```

```
/tmp/ipython-input-3213734980.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view
  new_df['tags'] = new_df['tags'].apply(stem)
```

```
new_df.head()
```

	movie_id	title	tags
0	19995	Avatar	in the 22nd century, a parapleg marin is dispa...
1	285	Pirates of the Caribbean: At World's End	captain barbossa, long believ to be dead, ha c...
2	206647	Spectre	a cryptic messag from bond’ past send him on a...
3	49026	The Dark Knight Rises	follow the death of district attorney harvey d...
4	49529	John Carter	john carter is a war-weary, former militari ca...

```
new_df['tags'][0]
```

```
'in the 22nd century, a parapleg marin is dispatch to the moon pandora on a uniqu mission, but becom torn between follow or
der and protect an alien civilization. action adventur fantasi sciencefict cultureclash futur spacewar spacecoloni societi
spacetravel futurist romanc space alien tribe alienplanet cgi marin soldier battl loveaffair antiwar powerrel mindandsoul 3
```

```
new_df['tags'][1]
```

```
'captain barbossa, long believ to be dead, ha come back to life and is head to the edg of the earth with will turner and el
izabeth swann. but noth is quit as it seems. adventur fantasi action ocean drugabus exoticisland eastindiatradingcompani lo
veofone'slif traitor shipwreck strongwoman ship allianc calypso afterlif fighter pirat swashbuckl aftercreditssting johnnyd
```

```
from sklearn.feature_extraction.text import CountVectorizer
cv = CountVectorizer(max_features=5000,stop_words='english')
```

```
vectors = cv.fit_transform(new_df['tags']).toarray()
```

```
vectors
```

```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]])
```

```
vectors[0]
```

```
array([0, 0, 0, ..., 0, 0, 0])
```

```
cv.get_feature_names_out()
```

```
array(['000', '007', '10', ..., 'zone', 'zoo', 'zooeydeschanel'],
      dtype=object)
```

```
ps.stem('dancing')
```

```
'danc'
```

```
stem('in the 22nd century, a parapleg marin is dispatch to the moon pandora on a uniqu mission, but becom torn between foll
```

```
'in the 22nd century, a parapleg marin is dispatch to the moon pandora on a uniqu mission, but becom torn between follow or
der and protect an alien civilization. action adventur fantasi sciencefict cultureclash futur spacewar spacecoloni societi
spacetravel futurist romanc space alien tribe alienplanet cgi marin soldier battl loveaffair antiwar powerrel mindandsoul 3
```

```
from sklearn.metrics.pairwise import cosine_similarity
```

```
similarity = cosine_similarity(vectors)
```

```
sorted(list(enumerate(similarity[0])),reverse=True,key=lambda x:x[1])[1:6]
```

```
[(1214, np.float64(0.28676966733820225)),
 (2405, np.float64(0.26901379342448517)),
 (3728, np.float64(0.2605130246476754)),
 (507, np.float64(0.255608593705383)),
 (539, np.float64(0.25038669783359574))]
```

```
similarity[0]
```

```
array([1.        , 0.08346223, 0.0860309 , ..., 0.04499213, 0.        ,
       0.        ])
```

```
def recommend (movie):
  movies_index = new_df[new_df['title'] == movie].index[0]
  distances = similarity[movies_index]
  movies_list = sorted(list(enumerate(distances)),reverse=True,key=lambda x:x[1])[1:6]

  for i in movies_list:
    print(new_df.iloc[i[0]].title)
```

```
recommend('Avatar')
```

```
Aliens vs Predator: Requiem
Aliens
Falcon Rising
Independence Day
Titan A.E.
```

```
new_df.iloc[1216]
```

	1216
movie_id	10641
title	Autumn in New York
tags	autumn in new york follow the sexual exploit o...

dtype: object

```
recommend('Gandhi')
```

```
Gandhi, My Father
Guiana 1838
The Wind That Shakes the Barley
Mr. Turner
A Passage to India
```

```
new_df.iloc[3647]
```

	3648
movie_id	38428
title	Gandhi, My Father
tags	with gandhi my father, produc anil kapoor and ...

dtype: object